# GestVox: Controlling a Robot with Voice-Specified Distance and Hand-Gesture Direction

Multimodal Human–Robot Interaction using Voice and Gesture Recognition

Course Project Report

**Machine Learning for Signal Processing (MLSP) – Project Based Learning**


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## Abstract

GestVox is a multimodal human–robot interaction prototype in which the *voice* says how far to move ("three") and a *hand gesture* seen by a webcam says which way (swipe left/right/up/down). A simulated robot on a $9 \times 9$ grid executes the command. Signal-processing chains for voice, surface EMG and camera video feed SVM, random-forest and $k$-NN classifiers, and a *confidence gate* rejects the command when either input is unsure. Voice recognition reaches 93.6 % accuracy, the joint command 78.0 %, and a live web demo reports measured statistics for every run.

# 1 Introduction and Objectives

Voice and gesture are the two most natural ways to command a robot, but each fails differently: speech in noise, gestures under poor lighting or occlusion. Early systems showed that combining them is more powerful than either alone [1], and multimodal interfaces can be more robust because one channel compensates for the other [2]. In GestVox the two channels fill *different slots of one command* instead of voting on the same label: voice $\rightarrow$ number of cells $N \in \{1, \ldots, 5\}$, gesture $\rightarrow$ direction $d \in \{\text{left, right, up, down}\}$. A wrong robot move is costlier than a repeated command, so a confidence gate rejects uncertain commands.

**Objectives:** (i) use public datasets [3], [4]; (ii) implement at least two signal-processing techniques; (iii) compare at least two ML algorithms with accuracy, precision, recall and F1-score; (iv) propose and evaluate confidence-gated complementary fusion; (v) build a working live prototype.

# 2 Theory and Literature Survey

## 2.1 Speech and EMG signal processing

Speech is analysed in 20–30 ms frames. A pre-emphasis filter flattens its spectral tilt,

$$y[n] = x[n] - \alpha x[n-1], \quad \alpha = 0.97, \tag{2.1}$$

and mel-frequency cepstral coefficients (MFCC) [5] summarise the spectrum on the perceptual mel scale,

$$m = 2595 \log_{10}(1 + f/700), \qquad c_k = \sum_{m=1}^{M} \log E_m \cos\Big[\tfrac{\pi k}{M}\big(m - \tfrac{1}{2}\big)\Big], \tag{2.2}$$

where $E_m$ are mel filterbank energies and $f$ is in Hz. Surface EMG measures forearm muscle activity (useful band 10–500 Hz, 50 Hz mains interference); filtering followed by time-domain features (MAV, RMS, waveform length) and spectral features (mean frequency) is standard [6], [7].

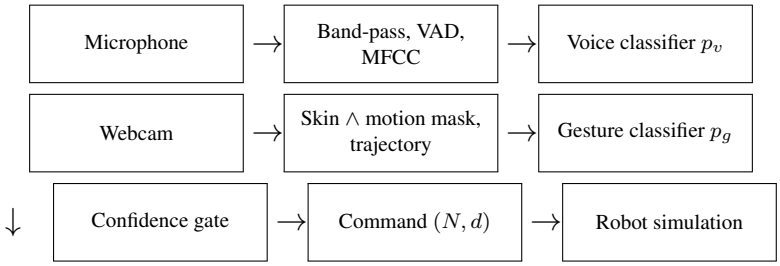



**Fig. 3.1** Block diagram of GestVox.

**Table 3.1** Datasets.

| Modality | Dataset and use | Source |
|---|---|---|
| Voice | Google Speech Commands v0.02 [3]: words *one–five*, 4 000 clips, 3 000 train / 1 000 test | `download.tensorflow.org` |
| EMG | UCI EMG data for gestures [4]: 8 channels, 200 Hz, 36 subjects; 4 gestures → directions; subjects 1–26 train (1 831 windows), 27–36 test (686) | `archive.ics.uci.edu/dataset/481` |
| Camera | Live OpenCV clips; initial model trained on 600 synthetic hand clips, user clips can be added | generated by the project |

## 2.2 Vision, classifiers and fusion

Gesture recognition extracts the hand, tracks it and classifies its motion [6]. A Haar-cascade face detector [8] removes the face; YCrCb thresholding gives a skin mask [9]; contours, convex hull and frame differencing come from OpenCV [10]. The classifiers compared are the support vector machine (RBF kernel) [11], random forest [12] and $k$-nearest neighbours [13], used through scikit-learn [14]. Multimodal fusion can be early, late or hybrid [15], and a classifier that may *reject* uncertain samples trades coverage for lower error among accepted ones [16]; GestVox applies this reject option to a two-modality command.

# 3 System Working and Methodology

## 3.1 Architecture and datasets

Fig. 3.1 shows the system. Table 3.1 lists the public datasets used.

## 3.2 Signal-processing chains

**Voice:** pre-emphasis (2.1), zero-phase Butterworth band-pass 80–4000 Hz, energy-based voice-activity detection, then 13 MFCCs (2.2) with deltas, cepstral mean normalisation and pooling over four time segments (130 features). **EMG:** Butterworth band-pass 10–90 Hz and 50 Hz notch, 1 s windows, MAV, RMS, waveform length, zero crossings, slope-sign changes and Welch mean/median frequency per channel (64 features). **Camera:** face masking, YCrCb skin mask with morphology, and a frame-difference motion mask so that *hand* = *skin* ∧ *moving*, which removes the static face and arm; the centroid trajectory gives net displacement $(\Delta x, \Delta y)$, path length and motion energy (10 features).

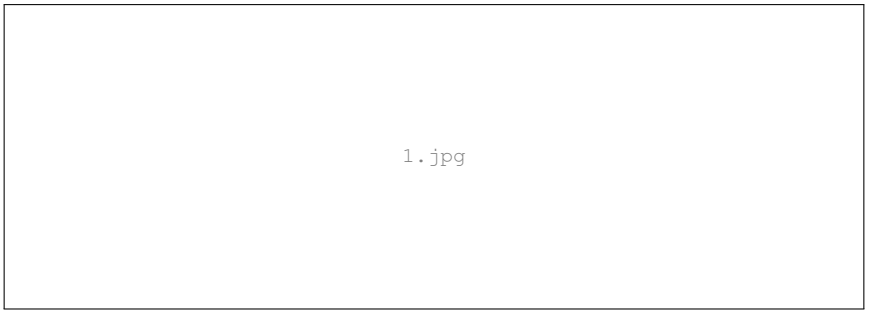



**Fig. 4.1** GestVox interface: controls and measured statistics (left), robot simulation (right).

### 3.3 Models and confidence-gated fusion

SVM (RBF, $C$=10, calibrated), random forest (300 trees) and $k$-NN ($k$=7) are trained for each modality behind a standard scaler. With posteriors $p_v$ and $p_g$, the command is accepted only if speech and a hand were detected and

$$\max_i p_v(i) \geq \tau \;\text{ and }\; \max_j p_g(j) \geq \tau \;\Rightarrow\; (N, d) = (\arg\max p_v,\; \arg\max p_g), \tag{3.1}$$

otherwise the robot does not move ($\tau = 0.45$ by default). A joint command is correct only if both slots are correct.

## 4 Implementation and Prototype

The software is written in Python 3.12 with NumPy and SciPy [17] for filtering, scikit-learn [14] for models, OpenCV 4.11 [10] for vision, and Streamlit [18] with Plotly for the web interface; only a laptop webcam and microphone are needed. A small browser component asks for camera and microphone permission and records for at most 2 s, stopping automatically 0.45 s after the spoken word ends (never before 1.2 s, so the swipe is captured). Audio (16 kHz) and frames (10 Hz, $160 \times 120$) go to Python, which shows *measured* statistics (level, SNR, hand-detection rate, trajectory, probabilities, stage times; about 0.2–0.4 s in total) and moves a $9 \times 9$ robot one cell at a time (Figs. 4.1 and 4.2).

## 5 Results and Discussion

Table 5.1 compares the models on held-out data (macro-averaged; EMG split is subject-independent; camera clips are synthetic, hence optimistic for real hands). The SVM is best for voice (93.6 %), the random forest for EMG (81.3 %). Removing the voice filtering chain lowers accuracy from 93.6 to 87.2 % (SVM), 91.6 to 82.6 % (RF) and 90.9 to 81.0 % ($k$-NN), so the DSP chain adds 6–10 points; on EMG filtering changed accuracy by only about $\pm 2$ points.

Joint accuracy (units *and* direction correct, random voice–EMG pairing) is 78.0 % on clean data and falls with noise (Table 5.2). The gate trades coverage for accuracy: at $\tau = 0.5$ it executes 94.1 % of commands at 80.4 % accuracy; at $\tau = 0.7$ it executes 62.1 % at 88.7 % accuracy and avoids 68 % of the wrong moves (Fig. 5.1).

Because a complementary command needs both slots, its accuracy is limited by the weaker




(a) result: units, direction, probabilities




(b) OpenCV: contour, hull, skin mask




(c) voice: waveform, classified window




(d) robot after a multi-cell move

**Fig. 4.2** Prototype screens: (a) command result, (b) gesture analysis, (c) voice analysis, (d) robot simulation.

**Table 5.1** Test accuracy / F1-score of the compared models.

| **Model** | **Voice (5 cls)** | **EMG (4 cls)** | **Camera (4 cls)** |
|---|---|---|---|
| SVM (RBF) | **0.936** / 0.937 | 0.789 / 0.787 | 0.967 / 0.967 |
| Random forest | 0.916 / 0.916 | **0.813** / 0.813 | **0.980** / 0.980 |
| $k$-NN | 0.909 / 0.909 | 0.774 / 0.773 | **0.980** / 0.980 |

**Table 5.2** Joint command accuracy (%) versus SNR with noise on one modality.

| **Noise on** | **Clean** | **20 dB** | **10 dB** | **5 dB** | **0 dB** |
|---|---|---|---|---|---|
| Voice | 78.0 | 75.7 | 53.7 | 28.1 | 19.4 |
| EMG | 78.0 | 76.7 | 59.3 | 37.6 | 26.9 |

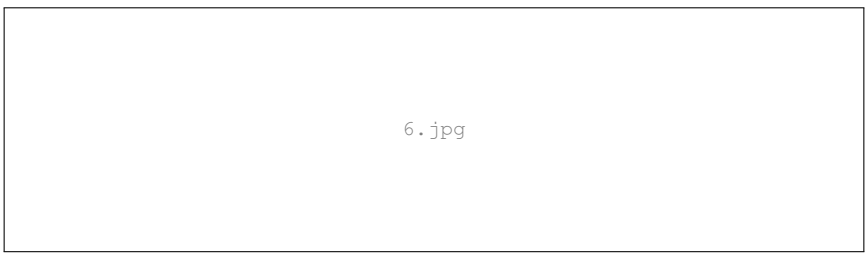



**Fig. 5.1** Joint study: accuracy under noise and gate-threshold sweep.

modality; the gate converts many errors into harmless rejections. In live tests voice worked reliably, while the camera gesture depended on lighting and a clear, fast swipe.

# 6 Conclusion and Future Work

GestVox shows that splitting a command between voice (distance) and gesture (direction), with a confidence gate, gives a simple and explainable multimodal interface; signal processing was essential for voice, three classifiers were compared on public datasets, and a live prototype re-

ports measured statistics. Learning outcomes: filter and feature design, classical computer vision, fair model comparison, fusion with a reject option and real-time web deployment. Touch-free control is relevant to assistive devices, collaborative and warehouse robots, sterile environments and smart homes. Limitations: simulated robot, random voice–EMG pairing, a camera classifier initialised on synthetic hands and lighting dependence. Future work: learned hand landmarks, a paired voice–gesture dataset, a larger vocabulary and a physical robot.

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